

1 Input Functions Relating to Batches

1.1 Basic screen

Basic terminal screen when a machine is assigned in batch mode:

The screenshot displays the AIP (Automated Input Process) terminal interface. It features a blue header with the AIP logo. The main content is divided into three sections: 'Machines/workplace', 'Operations on workplace', and 'Registered input materials'. The 'Machines/workplace' section shows a table with columns for Machine, Workplace, Status, Status since, and Note. The 'Operations on workplace' section shows a table with columns for Article, Order, Operation, Target qty., Yield, Scrap, N, and T. The 'Registered input materials' section shows a table with columns for Position, Batch, Material, Material designation, and Remaining. The bottom of the screen has a status bar with the mpcv logo, version information (CTAIP V2.0.2.67), server details (Server:win2008-10-3 TNR:999), and a timestamp (26.03.13 12:23:25).

Machine/	Workplace	Status	Status since	Note
CPL	Continue Pickling Line	PRODUCTION	26.03.13 08:00	
CRM	Cold Rolling Mill	PRODUCTION	26.03.13 08:18	
CEL	Electrolytic Cleaning Line	NO JOB	26.03.13 08:00	
CAF	BAF	NO JOB	26.03.13 08:00	

Article	Order	Operation	Target qty.	Yield	Scrap	N	T

Position	Batch	Material	Material designation	Remaining
01	R835-022	A06A2100926	2.10MM X 926MM HR COIL - SAE1006(BS	18 mt

The basic display shows the third list "Input materials currently logged on" for machines for which the "batch management" option is configured. This list shows all active input batches of the selected machine.

1.2 Order postings for operations subject to batch management

OP logon with input batches

By clicking the "log OP on" button a workflow including two tabs is opened. The operation to be logged on is selected in the first tab "select operation".

The "log operation on" tab is reached by clicking the "Next" button where in addition to the selected OP, the defined material components are displayed in a list.

By entering a batch number in the "input batch" field and clicking the "report batch" function a batch may be logged on as input material for a component. During the entry process, the terminal checks whether or not the batch number is known in the system and may be logged on. This is described in detail within the "input batch change" section.

"Batch" field

When an OP is logged on, a batch number is created simultaneously for the next output batch to be produced. The batch number may be assigned automatically or manually (please also see machine master settings --> MPL tab). The generated batch is created with the batch number in the system and set to the "running" status.

Provided that all required input materials are logged on, the OP may be started via the "OK" button in the "OP logon" dialog. Whether input material has to be logged on or not, can be defined in the assigned material type of the component.

Once the OP has been logged on successfully, all active input batches of the selected machine are displayed in the material list.

As of HYDRA-MPL product version 7.2.5:

If batches are logged on along with the OP and the user cancels the process or cannot log the OP on due to a plausibility check the input batches will be logged off automatically for this OP. In this case, batches are always logged off without indicating the consumption quantity.

By way of the following warning message, the user may confirm the logoff process:

The function that logs input batches off automatically can be activated/deactivated by a button in the hytnrcfg.ini file:

HYTNRCFG.INI
[MPL-Options 0/2xxx] ForceAutoLogOffInputBatches=0

Please note: This is only available as of HYDRA-MPL product version 7.2.5 and CTAIP version 2.0.2.3.

Logoff/interruption of OPs

A running OP may be interrupted or logged off by clicking the "logoff/interrupt OP" button. Then a dialog opens, where the following selection can be made:

If "log OP off" is clicked the logoff dialog opens that contains the same input fields like the "output batch change" dialog (see next section).

Thus, the output batch that is currently still active is completed, when OPs are interrupted or logged off.

1.3 Postings based on batches

Input batch change

Input material can be changed for a running OP if the "input batch change" button is clicked.

Log input batch off

Input batches may be changed by entering a currently active batch number or by entering a new batch number. When logging batches off, it is also possible to enter the status and consumption of the batch to be logged off.

Options when logging input batches off:

F1 - PROCESSED

The batch is set to the "processed" status and the remaining quantity that is still available is set to 0. A consumption posting is generated as goods issue for the current, remaining quantity.

F2 - BLOCKED

The batch is set to the "blocked" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue.

F3 - with remaining quantity

The batch is set to the "free" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue. If the remaining quantity that is still available becomes ≤ 0 , the batch status automatically switches to "processed".

Consumption

The entered consumption (unit of the input material) is deducted from the remainder of the batch and a goods movement is generated.

Comment on batch

The entered comment is saved as information for the batch.

Log input batch on

Provided that the batch is known, batch data are displayed in an intermediate dialog where the logon may be confirmed.

Provided that the batch could be logged on, it is taken over to the material list in "customer batch number" and thus the change is completed.

However, in case the logon is inadmissible as the input material does not correspond to that of the component, the logon is rejected by the following error message:

Logon of unplanned input material

In addition to planned materials, it is also possible to log "unplanned" material on for an OP, using an additional feature at the component of the OP. If the "replaceable" option is set to "J" the user is able to assign the respective component manually, when batches are logged on. However, the logon is only allowed if the material type of the input batch corresponds to that of the component.

Within the selection list the components are filtered according to the material number and displayed as follows:

Logon of unknown batches

An input batch, which is not yet known in HYDRA, may be logged on for an OP using the "creating unknown batches" option in the basic settings of HYDRA (in BDE --> settings 2 tab):

In this case, it is searched for a valid assignment of input material to the material type of the selected component, when input batches are logged on. Provided that a corresponding assignment is found and the "allow entry of unknown input batches" option is configured in the "input batch processing" tab at the material type, the batch is generated by logging it on to the system and set to the "running" status at the same time. The batch is initially created in a quantity of 1.000.000.000.

Output batch change

Output material may be changed for a running OP using the "output batch change" button.

The input batches that are logged on are displayed with their available remaining quantity in a list within the output batch change dialog. The following data may be entered:

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here. The selection refers to transport units that were assigned to the material type of the OP.

Comment on batch

In this field a comment may be saved for the batch to be produced.

Quantity

The batch to be produced is posted with the quantity entered here. The quantity is taken over as primary quantity of entry to the order and machine and a goods movement is generated as goods receipt.

Quality

A batch may be classified as yield or scrap quantity. The system posts yield batches with the "free" status. Scrap batches automatically get the "blocked" status. When scrap is selected, a valid scrap reason has to be assigned.

"Preceding batches" function key:

This button opens a list with output batches which have already been produced for this OP.

"Change inp. batch" function key:

Using this button the user can switch to the "input batch change" function.

New batch

When a current output batch is completed, a new batch number is simultaneously created for the next batch. The batch number may be assigned automatically or manually. The batch generated in this way is created with the batch number in the system and set to the "running" status.

Entry of additional batch attributes

Several additional batch attributes may be recorded for a material type by the configuration MPL --> Master data --> Attributes.

In case attributes are defined to be recorded at the terminal for the material type of the running OP, another input dialog is opened, when the output batch is changed. The dialog opens additionally after clicking OK in the output batch change function, interrupt OP and finish OP function.

Example when two additional attributes are collected:

Using batch attributes, numeric and alphanumeric values may be recorded which are then saved for the produced batch in an additional table.

Enter goods receipt batch

A new goods receipt batch may be created in the system by the "enter GR batch" button.

Having clicked the "OK" button, the batch is created and the dialog remains open for further entries.

Batch numbers may be generated automatically or manually depending on the configuration. Moreover, the following data are saved at the batch.

Workplace

Machine where the batch was recorded

Operation

Order where the batch was recorded

Material

Material number of the batch

Quantity and unit

The batch is created with the quantity and unit entered here. A goods receipt is posted with this quantity.

Quality

A batch may be classified as yield or scrap quantity. The system creates yield batches with the "free" status. Scrap batches automatically get the "blocked" status and the batch class "scrap". When scrap is selected, a valid scrap reason relating to the corresponding workplace has to be assigned.

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here.

Comment on batch

A comment may be saved for the goods receipt batch in this field.

Repost batch

Using the "repost batch" button, an existing batch may be reposted to another material buffer.

Having clicked the "OK" button, the batch is rebooked to a new material buffer and the dialog remains open for further entries.

As an alternative, the batch can also be reposted from yield to scrap.

1.4 Throughput batch mode

A HYDRA machine may also be configured in "throughput batch mode". In throughput batch mode input material is continued being processed with unchanged number (throughput batch number) via an OP.

The entry functions for throughput batch processing at the terminal are identical to those for active batch tracing at the machine.

Please note/restriction: At machines with "throughput batch mode" it is impossible to log operations on at the same time.

1.5 Manual partial upload in HYDRA-MPL environment

Using the manual partial upload function of HYDRA-ADE (A_TR dialog), it is possible to record a partial quantity for the current operation and thus for the active output batch.

The following performance as regards the entry of quantities results from a manual partial upload/confirmation:

- Scrap is only transferred to the active operation and not to the output batch
- Yield is accumulated on the current output batch
- In this case, log records of the record type "H" are also generated when the shift changes, provided that yield has been recorded beforehand.
- In case yield has already been recorded as partial quantity for an output batch, it is no longer possible to log the batch off as scrap, when the output batch is changed the next time. In this case, the batch assigned to the "yield" class has first to be completed, before scrap can be posted.

Effects on available retrograde material components are as follows:

- When a total quantity is recorded, scrap is not deducted in a retrograde manner
- When yield/scrap is recorded, withdrawal also takes place when it comes to scrap
- Negative values, which might result from quantity offsetting, are not considered

The following notes and restrictions are to be taken into account:

- Personal partial uploads (with P_AB) are not taken into account

- Scrap quantities are only uploaded to a PPS system, provided that the interface has been configured for the “upload of partial confirmations”
- At the moment it is only supported to post quantities of the “yield” and “scrap” accounts.
- The partial upload itself does not trigger a goods movement. This is only the case, when the output batch has been completed.
- The checks relating to the input quantity and the display within the consumption balance cannot be used together with the "collect input quantity in relation to batches" configuration within the material type, as the current output batch is not completed for partial quantities. This affects, among other things, the collection of serial numbers with batches (MPL-SNR).

1.6 Display of produced output batches

Using the third list, it is possible to display the output batches produced for a running operation that is subject to batch management within the machine master at the terminal.

A maximum number of 20 output batches is displayed for each machine in the list. Yield and scrap batches are shown, which have been produced at this terminal since output batches were changed.

As this list is only kept locally by the respective terminal, it is not synchronized with the server, when AIP is started.

The list includes, among others, the article, article designation, batch number, date, time, quantity, batch class, user fields and alternative batch numbers. Individual fields might not be assigned values, which depends on the input scenario when generating output batches.

Configuration:

List contents can be configured via the [MNR_AMAT.LST] section in the ctaiply.ini file.

Example:

CTAIPLAY.INI
<pre>[MNR_AMAT.LST] GRID_FONT=Arial GRID_FONTSIZE=9 GRID_COLOR=clBlack GRID_BACKGROUND=clWhite GRID_ORDER=ROW.IDX=- GRID_CAPTION=produced output batches</pre>

```

EXAMINE_SCANEXPR1=KLASSE=G
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=KLASSE=A
EXAMINE_SCANCOLOR2=clRed

; ROW.IDX=N10,50,R,Row
CNR=C20,150,L,batch number
;CNR=*CNR,batch
ATK=C25,125,L,article
; ATK=*ATK
; KLASSE=C3,40,Z,*
MENGE=N12.0,70,R,quantity
EINH=C3,30,Z,ME
DAT=dd.mm.yyyy,70,L,date
ZEI=hh:mm:ss,60,L,time
ATKBEZ=C30,200,L,article designation

```

Display as third list:

Figure: Display of produced output batches



Using the  button, the user can switch to the display of output batches.

Server-based comparison of produced output batches

As of the AIP program version V# 2.0.2.54 the list of produced output batches of a machine can be synchronized with the server using a cyclic comparison function.

The server comparison [\[Technically: PDM list 13 with MOD=P\]](#) is enabled by the [configuration](#) within the customized configuration file "ctlisten.ini".

Output batches recorded within the last three days at this terminal are only compared.

1.7 Batch information

Batch information is displayed in a dialog, when the "batch info" button is clicked.

1.8 Display of consumption balance

When logging an OP off, the "consumption balance" (V_BLZ dialog) may be displayed, which has to be configured via the machine and the material type of the operation.

The consumption balance is shown, provided that this option has been activated at the machine and material type.

The consumption balance shows the material consumption based on batches and the user is able to log running batches off.

Figure: Consumption balance – V_BLZ

Configuration:

A special configuration (function = "DLG=V_BLZ;BREAK-ON-CANCEL") has to be defined for the OK button of the logoff dialog (e.g. A_AB_MPL) to make sure that the consumption balance is started, when the OP is logged off.

Sample configuration of the OK button for starting the consumption balance:

Figure: Example for the OK button configuration including consumption balance

Display of consumption postings

The "show details" function allows for the consumption quantities, which have been collected so far, to be displayed.

Figure: Display of consumption quantities – V_BLZ_DTL

Logging input batches off

By way of the "log batch off" function, the user can choose a currently running batch from the list and log it off by entering the consumption.

Figure: Log batch off – V_BLZ_CEAB

1.9 Advance logon of input batches

The AIP terminal provides the following dialogs or functions to log input batches on in advance. The function “advance logon of input batches” has been designed to be able to “set up” and log on the next input batch, while another OP and input batch are currently running. This next input batch is not yet active but assigned the flag “logged on in advance”.

An input batch can be logged on in advance for a currently running OP or for a prepared OP.

General

Usage

The process might require an input batch to be logged on in advance and set up accordingly on a machine, while the preceding input batch is still being used for a material.

This situation frequently occurs at very large machines processing, for example, roles or belts that are uncoiled as input batch at the beginning of the machine and coiled up as output batch at the end of the machine.

As the users are mostly busy with activities at the end of the machine at the time when the input batch actually needs to be changed, they cannot perform the input batch change and, as a result, they are provided with the opportunity to log the next input batch on already in advance for an order/OP.

Then the input batch can actually be changed by logging a new OP on or a project-specific call can be established.

Configuration

The settings required for using the function “advance logon of input batches” is described [here](#).

Process/procedure

The procedure for using the function “advance logon of input batches” or the logical process is described [here](#).

AIP usage

AIP basic screen

The basic AIP screen shows the function key “Advance logon of input batch” (preregistration of input batch). The dialog for logging input batches on in advance may be used by clicking this function key.

Advance logon of input batches (CE_VWL_MPL)

The user selects the workplace to which an input batch is to be logged on in advance in the basic screen. The below dialog (CE_VWL_MPL) opens by clicking the function key "Advance logon of input batch" (preregistration of input batch).

If an operation is currently running/logged on to the workplace, this one will be selected by default. The input batch (that is to be logged on in advance) is entered/scanned for the selected BOM item. Advance logon of input batches is started by clicking the button "post batch".

At first the input batch is checked for validity (dialog CE_VAN). The material number of the input batch is checked against the material number of the component list or the BOM item. The input batch is logged on in advance, once the button "log input batch on in advance" has been clicked.

Finally, the input batch that has been logged on in advance is displayed in purple in the BOM of the component. The dialog is exited by the "cancel" key.

Show third list

Besides the logged on/running input batches, the third list of the basic screen also shows the input batch that has been logged on in advance and highlights it in purple.

Consequently, a BOM item can use a logged on/running input batch and an input batch logged on in advance at the same time.

Log off batch logged on in advance

The dialog "advance logon of input batches" also provides the opportunity to log off or reset an input batch that has been logged on in advance.

To do so, the input batch that has been logged on in advance is selected once more and the user confirms the button "post batch".

The input batch is logged off by the dialog CE_VAB.

Log on input batch logged on in advance with OP

When logging a prepared operation on, an input batch that already has been logged on in advance is shown for a BOM item. The input batch that has been logged on in advance is logged on automatically by logging the operation on.

If, however, another input batch is logged on for this BOM item, the input batch logged on in advance is kept for the running OP.



When an OP is logged off/interrupted, all input batches logged on in advance are also logged off/reset automatically.

1.10 Recording of serial numbers

Serial numbers are assigned to be able to distinguish between individual items of material. Consequently, the combination of material number and serial number uniquely identifies an individual item.

MES provides several variants including different features to record serial numbers:

- Variant 1: Entry of serial numbers for OPs that are not subject to batch management (dialog A_SNR/“E” only)
- Variant 2: Entry of serial numbers for OPs that are subject to batch management (dialog A_SNR)
 - Manual input of the serial number. It is assigned to a HYDRA batch number.
 - Automatic assignment of the serial number = HYDRA batch number
 - Automatic assignment of the serial number. It is assigned to a HYDRA batch number

1.10.1 Entry of serial number for OPs that are not subject to batch management (dialog A_SNR/ “E”)

General/utilization

Serial numbers are assigned in ERP when creating a production order. The assigned serial numbers are transferred as detailed information for the order header where they are managed accordingly in the interface to HYDRA.

The OP of the order that is not subject to batch management is logged on and the relevant serial numbers are recorded with the relevant quality (yield/scrap).

The serial number is entered manually (e.g. also barcode for the individual item) by the user on AIP. To simplify data collection, the serial numbers that are still available for the order are shown in a list where they can be selected. Please note that a serial number can occur as scrap only once.

The quantity recorded for the order/OP is always 1 for each individual item and is posted onto the relevant yield or scrap account of the order/OP subject to the quality.

Uploads (goods movements) relating to serial numbers can be transferred to the ERP system for recorded serial numbers/individual items.

Configuration

The required settings for using the function “collection of serial numbers” (without orders that are subject to management in batches) are described [here](#).

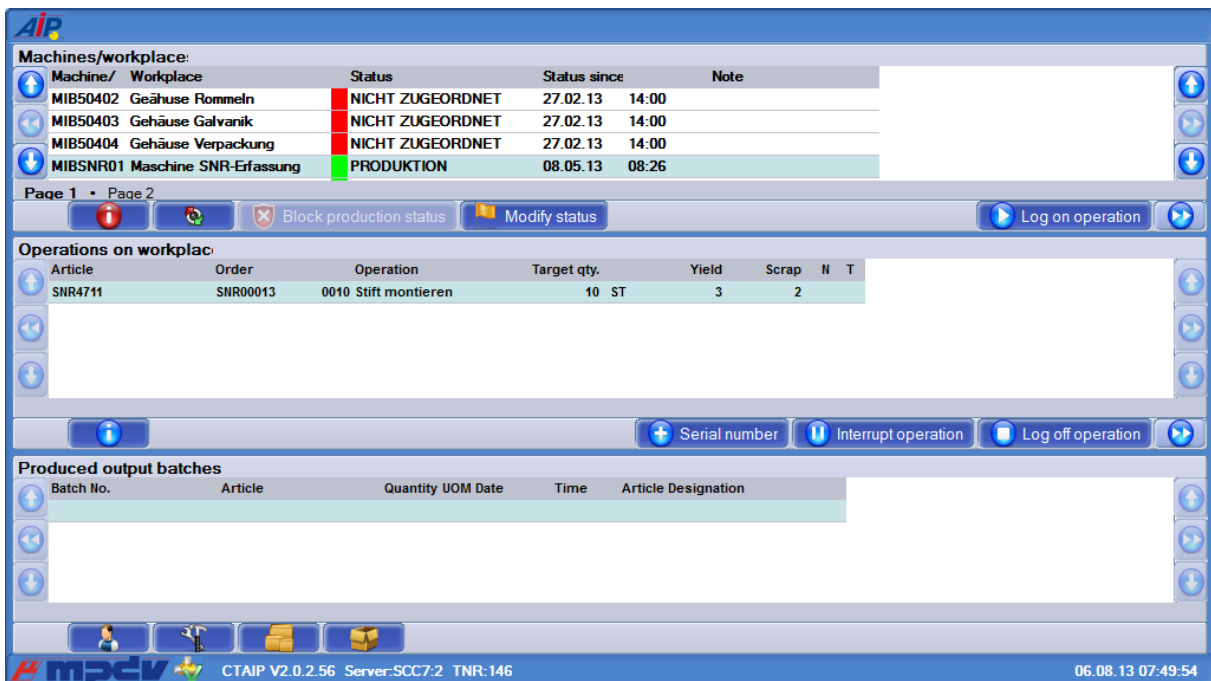
Procedure

The procedure for using the function “collection of serial numbers” (without orders that are subject to management in batches) or the logical process are described [here](#).

Utilization with AIP

Basic AIP screen

The operator uses the dialog “serial numbers” (A_SNR). The relevant function key is configured in the basic view of AIP.



The screenshot displays the AIP interface with the following sections:

- Machines/workplace:** A table showing machine status.

Machine/	Workplace	Status	Status since	Note
MIB50402	Gehäuse Rommeln	NICHT ZUGEORDNET	27.02.13	14:00
MIB50403	Gehäuse Galvanik	NICHT ZUGEORDNET	27.02.13	14:00
MIB50404	Gehäuse Verpackung	NICHT ZUGEORDNET	27.02.13	14:00
MIBSNR01	Maschine SNR-Erfassung	PRODUKTION	08.05.13	08:26
- Page 1 • Page 2** (Navigation buttons)
- Operations on workplace:** A table showing current operations.

Article	Order	Operation	Target qty.	Yield	Scrap	N	T
SNR4711	SNR00013	0010 Stift montieren	10 ST	3	2		
- Produced output batches:** A table for recording produced batches.

Batch No.	Article	Quantity	UOM	Date	Time	Article Designation
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At the bottom, there is a status bar with the mpciv logo, version information (CTAIP V2.0.2.56, Server:SCC7.2, TNR:146), and a timestamp (06.08.13 07:49:54).

Figure: Basic view including function key “serial number” (A_SNR)

Entry of serial numbers (A_SNR)

Figure: Entry of serial numbers on the terminal (A_SNR/ “E”)

Serial number

The user enters the serial number. The serial number has to be part of the order. A validation check takes place. If the serial number is not part of the order or if it already has a result (already recorded) it cannot be used.

By clicking , a selection list showing the “free” serial numbers for the order is displayed.

Requesting the selection list simultaneously updates the list of serial numbers that have already been recorded.

Grid

The displayed list shows all serial numbers that have already been entered including their relevant quality (yield/scrap).

- Scrap is shown red
- Yield is shown green

Quality

If a serial number is entered, it has to be assigned quality. The following can be chosen:

- Yield
The serial number is entered with the quality “yield”. The operation quantity is increased by yield = 1. The default value is “yield”.
- Scrap
The serial number is entered with the quality “scrap”. The operation quantity is increased by scrap quantity = 1.
- Scrap reason
If the quality is entered as scrap the user also has to enter or select a scrap reason.

“Capture” function

By clicking this function, the entered serial number is recorded with the selected quality.

“List” function

By clicking this function, the entire list of serial numbers that have already been entered including the selected quality can be updated (e.g. if the dialog is opened anew).

Interrupt/finish OP for orders with serial numbers (A_UN/ A_AB)

Quantities are not entered when operations subject to management in serial numbers are logged off. The quantity fields have to be disabled in these dialogs.

1.10.2 Entry of serial numbers for OPs subject to management in batches (dialog A_SNR)

General/utilization

If an order/operation subject to management in batches is used to enter serial numbers, there are the following alternatives to enter or assign the serial numbers for data collection:

- Manual input of the serial number. It is assigned to a HYDRA batch number.
- Automatic assignment of the serial number = HYDRA batch number
- Automatic assignment of the serial number. It is assigned a HYDRA batch number

Configuration

The required settings to use the function “collection of serial numbers” (without orders that are subject to management in batches) are described [here](#).

Procedure

The procedure to be able to use the function “collection of serial numbers” (without orders that are subject to management in batches) or the logical process are described [here](#).

AIP usage

Entry of serial numbers (A_SNR) including manual assignment of serial numbers – characteristic “E”

If batch management requirement is enabled for the operation, a batch is created additionally for each serial number in the material and production logistic.

Consequently, the relation between the current output batch and the serial number posted in the database are saved for tracing. In this case, the output batch is considered as ID without quantity and, as a result, it does not have a quantity.

Batch attributes that might be required to be recorded for OPs handled in batches are recorded by way of the commonly used standard dialog before sending.

The “collection of serial numbers” dialog is structured as described below. The serial number can be entered. Batch assignment is performed in the background.

Entry of serial numbers (A_SNR) including automatic assignment of serial numbers – characteristic “G”

By using the option “serial number requirement” = “G” for the operation, output batches are entered as serial number. In this case, the serial number is the output batch number.

The field “serial number” is disabled and includes the current output batch. Posting is possible, once the badge number has been entered by the function “capture”. An applicable scrap reason has to be entered, provided that a “quality indicator” unequal to “yield” has been chosen.

Exactly one batch with quantity 1 is created for each serial number and the serial number matches the batch number in this case.

The function “list” updates the list of serial numbers that have already been recorded.

Batch attributes that might be required to be entered for OPs handled in batches are recorded by the common standard dialog before sending.

Entry of serial numbers (A_SNR) including automatic assignment of serial numbers by the number range - characteristic - "S"

Serial numbers in relation to a batch are recorded by the option "serial number requirement" = "S" at the operation. For this purpose, the operation needs to be subject to batch management. An output batch change (CA_WL) is triggered every time a serial number is posted. This output batch change creates a batch with quantity 1 for the serial number in MPL.

By clicking the function  a new serial number is determined on the server and shown in the field "serial number". The new serial number is assigned uniquely for the entire system by the number range SNR.

Batch attributes that might be required to be entered for OPs handled in batches are recorded by using the standard dialog before sending.

Interrupt/finish operation for orders including serial number tracking

Quantities are not entered when operations subject to management in serial numbers are logged off. The quantity fields are to be disabled in these dialogs.